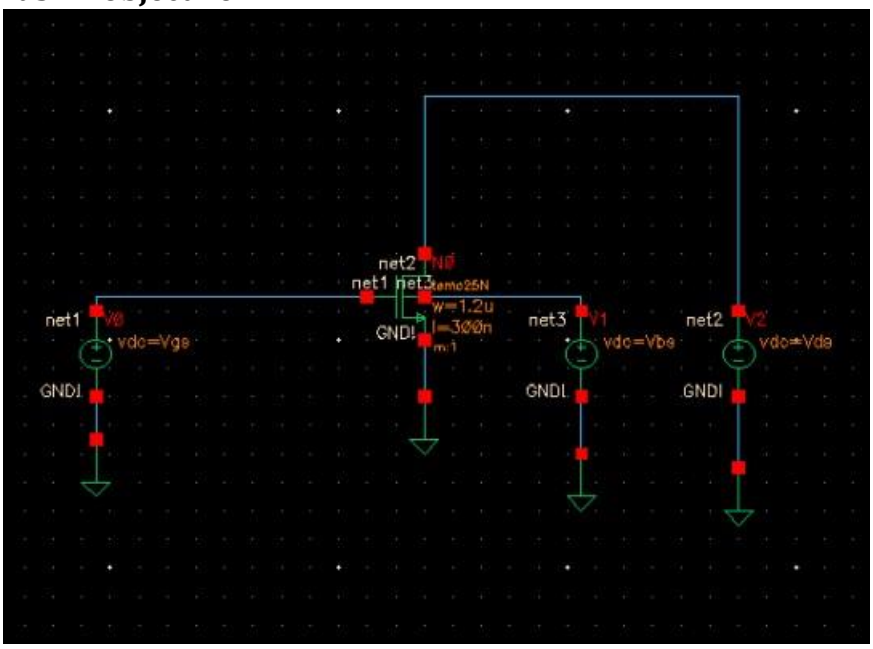
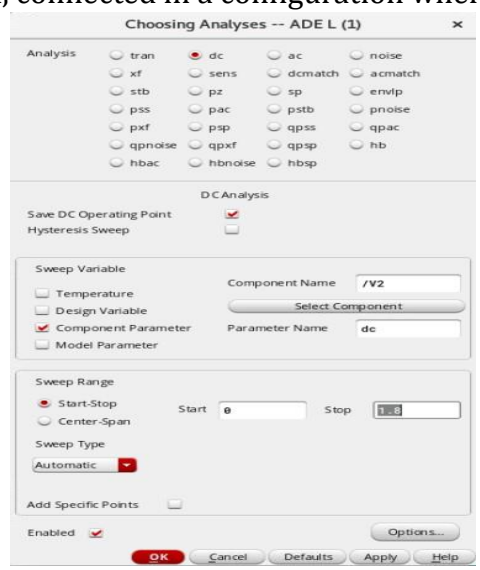
Objective: Brief overview of the lab, objectives

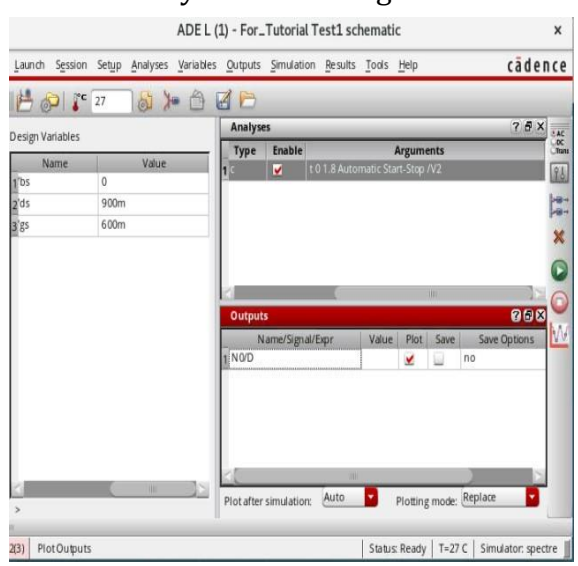
Followed by each section outlined in the lab

1. **Task 1 objective**

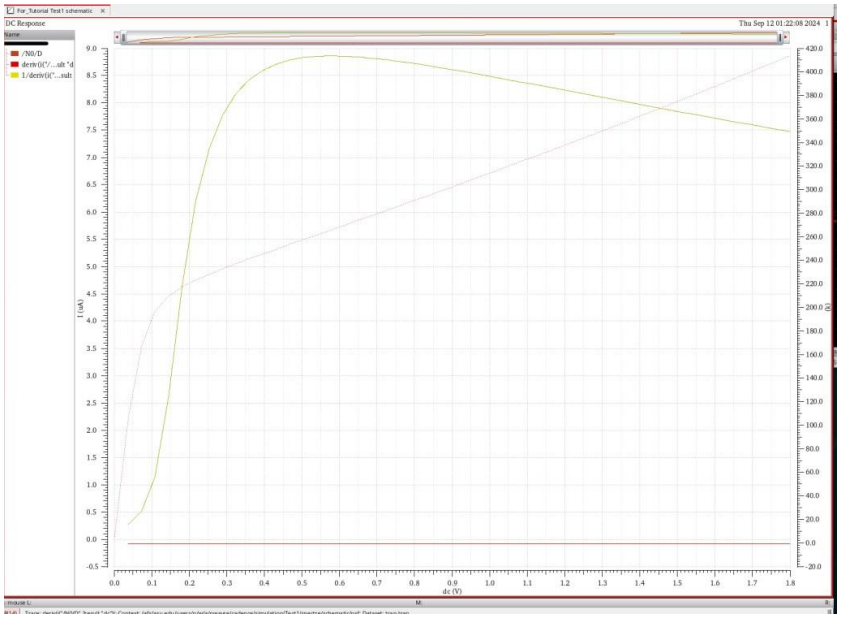


The schematic depicts an NMOS transistor with dimensions W=1.2µm, and L= 300nm, connected in a configuration where Vg and the bulk by Vb drive the gate.





Above images show the ADL Setup



ID vs VDS graph

a) The above graph shows the ID - VDS relation for the circuit and the estimated ro(output resistance) for the circuit is

Ro = 402.3 k ohms

b) After adjusting the gate length to 2um and the same W/L = 20

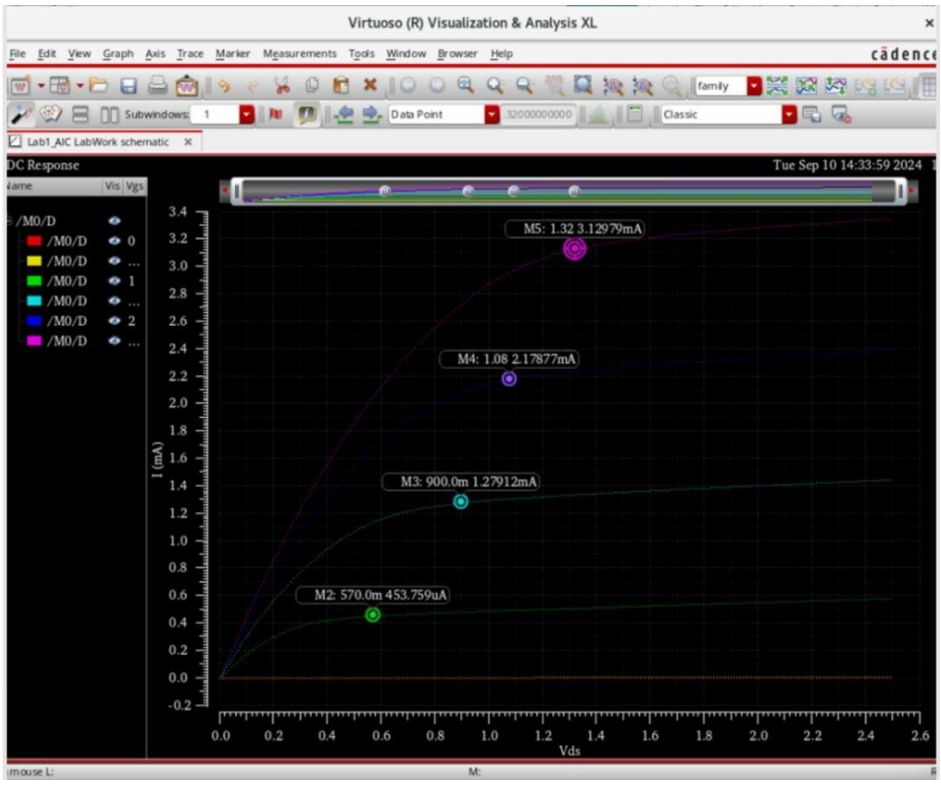
Ro = 132 k ohms

c) Fixing VDS at 1.5V and sweeping VGS voltage

Transconductance (g) = 3.282 A/V

And,

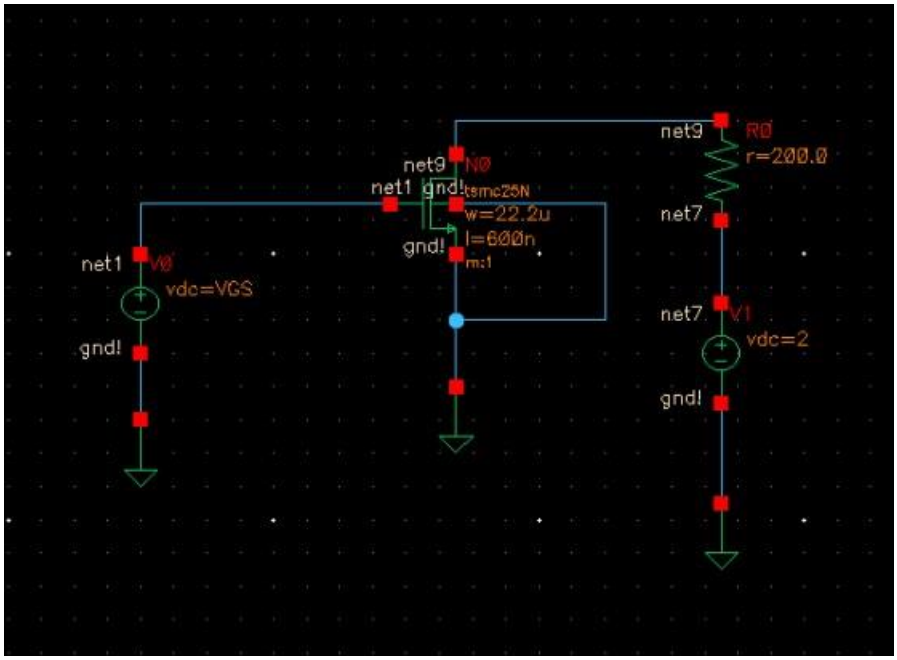
Vth = 422.3 mV



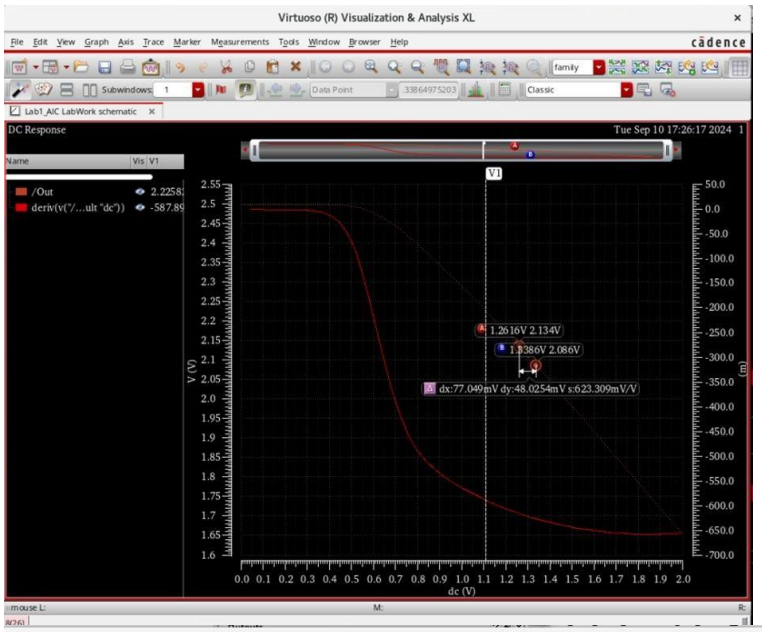
Parametric sweep

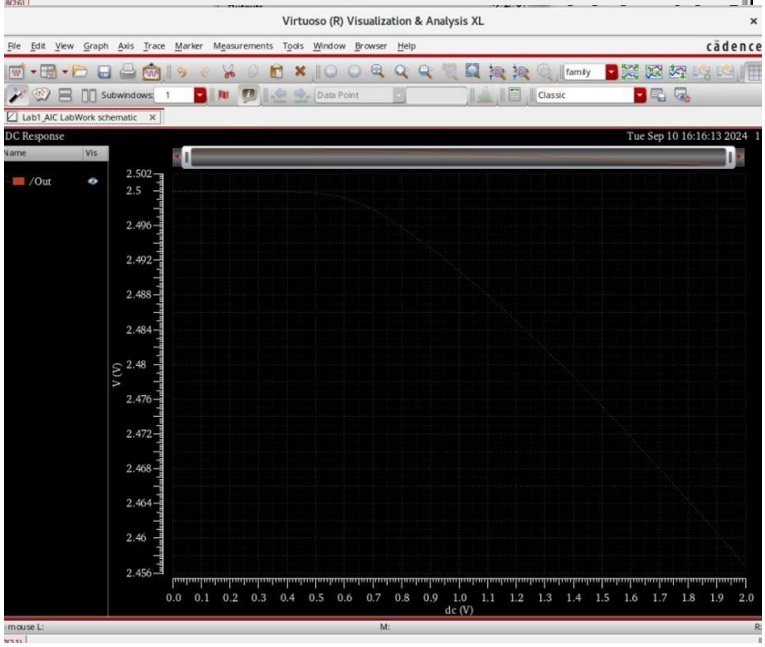
Task 2 – Objective

Q2 Schematic : NMOS



Objective : to calculate the value of W/L in the following circuit


Virtuoso (R) Visualization & Analysis XL
File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help
cadence
Lab1_AIC LabWork schematic
DC Response
Tue Sep 10 17:26:17 2024
/Out
deriv(v("/...ult "dc"))
1.2616V 2.134V
1.3386V 2.086V
dx:77.049mV dy:-48.0254mV s:623.309mV/V
V (V)
dc (V)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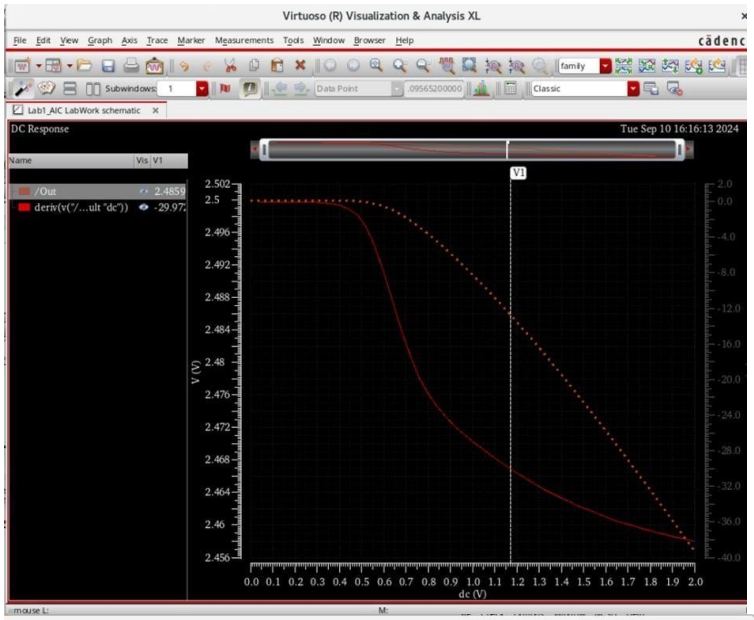
cadence
Lab1_AIC LabWork schematic
DC Response
Tue Sep 10 16:16:13 2024
/Out
V (V)
dc (V)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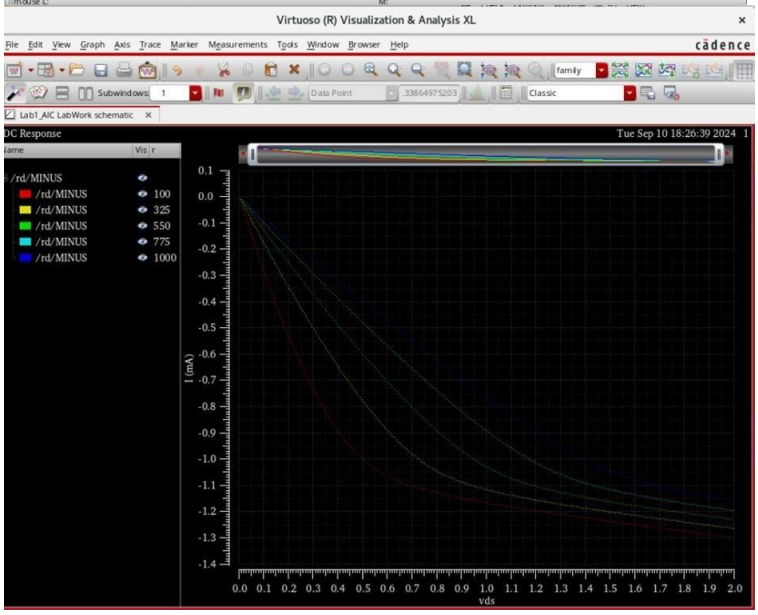
Data Point
Classic
Lab1_AIC LabWork schematic
DC Response
Tue Sep 10 16:16:13 2024
Name
Vis
V1
/Out
2.4859
deriv(v("/...ult "dc"))
-29.97
V (V)
dc (V)


Virtuoso (R) Visualization & Analysis XL
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cadence
Subwindows: 1
Data Point
Classic
Lab1_AIC LabWork schematic
DC Response
Tue Sep 10 18:26:39 2024
/rd/MINUS
/rd/MINUS 100
/rd/MINUS 325
/rd/MINUS 550
/rd/MINUS 775
/rd/MINUS 1000
I (mA)
vds

Q.2) b)

$$V_d = V_{dd} - I_D R_D$$

$$V_d = 2 - (200\mu)(0.2K)$$

$$\boxed{V_d = 1.96V}$$

$\therefore$ Nmos is in Satusation region

Now,

$$g_m = \frac{2I_d}{V_{OV}}$$

$$\therefore \quad g_m = \frac{2 \times 200 \times 10^{-6}}{0.2} = 0.002 \, A/V$$

$$\boxed{g_m = 2 \, mA/V}$$

Now Small Signal gain $A_V = -g_m R_D$

$$\boxed{A_V = -0.4}$$

2.

Q.2) b)

Given : $V_{ov} = 0.2V$

$I_d = 200 \mu A$

To find : w, g_m, A_v

$$I_D = \frac{1}{2} \mu C_{ox} \cdot \frac{w}{L} (V_{gs} - V_{tn})^2$$

$$200 \times 10^{-6} = \frac{0.944 \times 10^{-4}}{2} \left(\frac{w}{L}\right) \times (0.2)^2$$

Let $L = 300n$

$$\frac{w}{L} = 105.93$$

$$w = 105.93 \times 300 \times 10^{-9}$$

$$\boxed{w = 31.7 \mu m}$$

Now, $g_m = 2ms$

& $A_v = g_m . R_D$

For Pmos in Sat:

$$R_D = \frac{V_{DD} - V_{DS}}{I_D}$$

$$\boxed{R = 9K\Omega}$$

& $A_v = -2ms \times 9K$
$= -18$

Task 3 Q3 objective find W/L for 3 ma of drain current assume Vov = 0.4, also fond values and plot for other following questions

Q:3) a)

Given : $I_D = 1 \times 10^{-3} A$

$R_D = 1 \times 10^{3} \Omega$

$\therefore V_D = 1V \qquad (V_D = I_D R_D)$

$\therefore V_{ov} = 0.2$

$V_{ov} < V_D$

$\therefore$ Transistor is in Saturation region

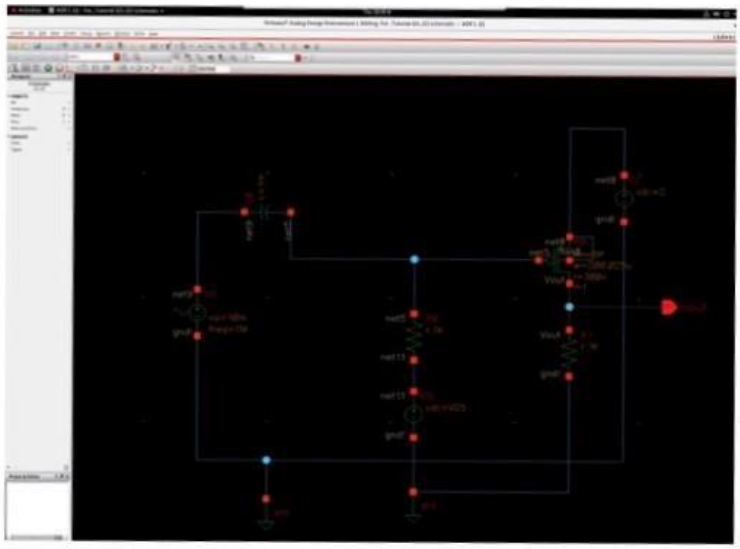
Now,

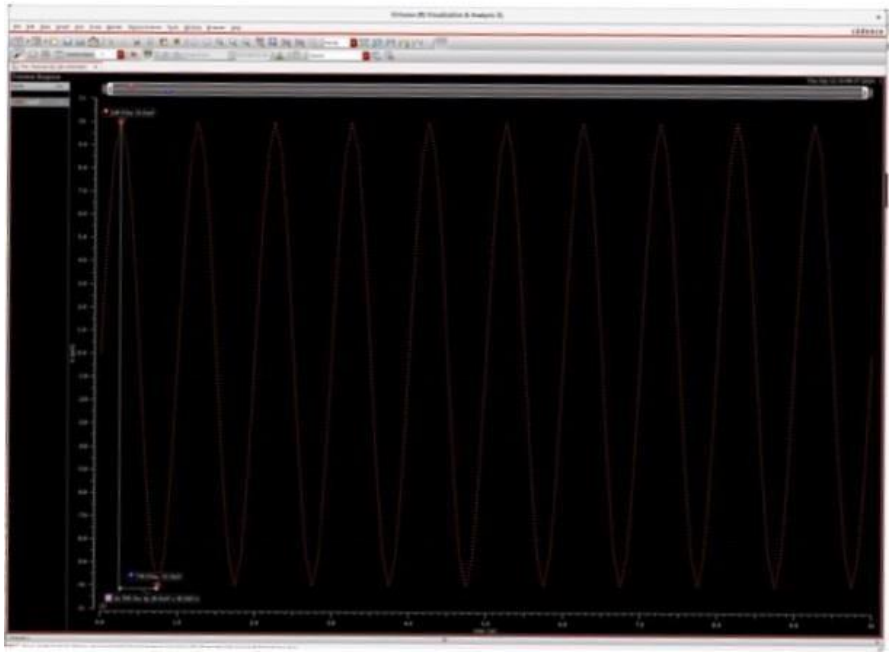
$$g_m = \frac{2 \cdot I_D}{V_{ov}}$$

$$\boxed{g_m = 0.01 \ A/V^2}$$

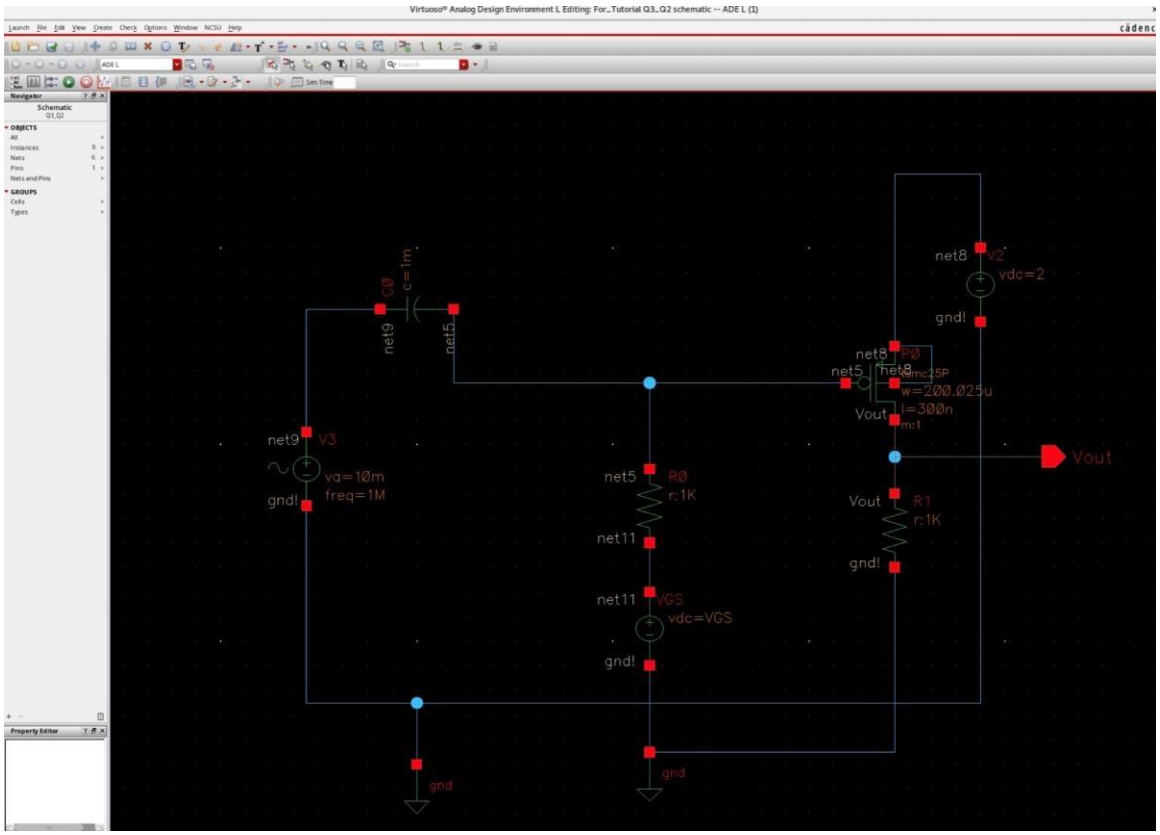
$$A_v = -g_m R_D = -0.01 \times 1K$$

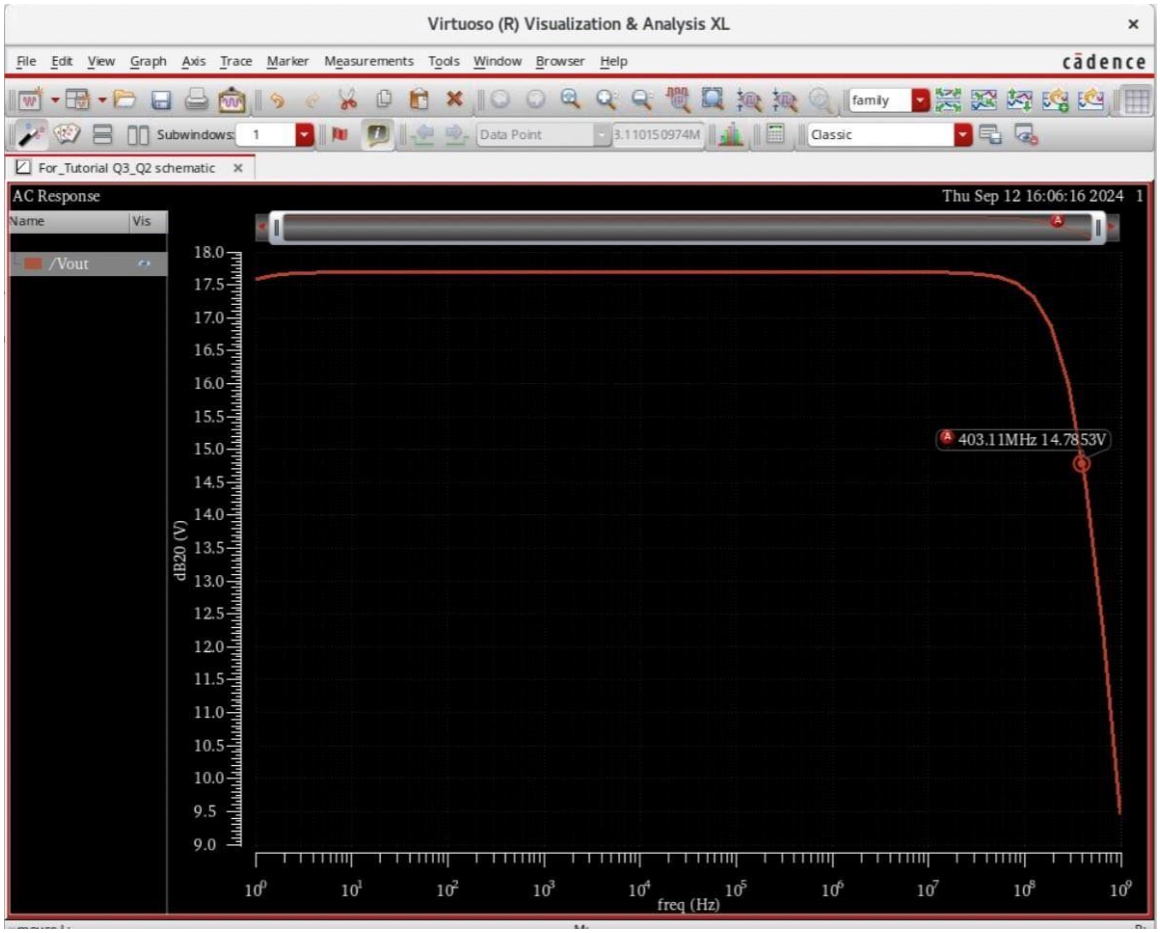
$$\boxed{A_v = -10}$$





3.


Virtuoso® Analog Design Environment L Editing: For_Tutorial Q3_Q2 schematic -- ADE L (1)
C0
c=1m
net9
net5
net8
V2
vdc=2
gnd!
net8
P0
net5
net8
nmc25P
w=200.025u
l=300n
m:1
Vout
net9
V3
va=10m
freq=1M
gnd!
net5
R0
r:1K
net11
net11
VGS
vdc=VGS
gnd!
Vout
R1
r:1K
gnd!
Vout
gnd
gnd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family
Subwindows: 1
Data Point
3.110150974M
Classic
For_Tutorial Q3_Q2 schematic
AC Response
Thu Sep 12 16:06:16 2024 1
Name
Vis
/Vout
403.11MHz 14.7853V
dB20 (V)
freq (Hz)